ADITYA MEHRA — ROBLOX

Applied Scientist - AI & World Simulation Intern (2027) | Dossier APX-AI-006 | Comp: $64/hr | Bengaluru • +91-7003456624 • ashishiash007@gmail.com

QUANTITATIVE THESIS: Spatial Graph Reasoning, Real-Time Distributed Telemetry & Generative Physics State Engines

Designing scalable event synchronization fabrics for multi-agent simulation environments under massive player concurrency.

Strategic Engineering & Operations Qualifications

- **Autonomous AI Systems Architecture:** Engineered deterministic multi-agent execution DAGs, automated verification gates, and evaluation fixtures.
- **High-Throughput Data Infrastructure:** Built sub-millisecond SQLite FTS5 search engines coordinating 9,816 cross-repository entities across 7 domains.
- **International Business & Quantitative Rigor:** BBA International Business (Class of 2026, CGPA 8.78/10.0) combining macro financial literacy with distributed backend engineering.
- **Production Reliability & Verified PRs:** Authored verified upstream pull requests resolving compiler defects in enterprise TypeScript and Python codebases.

Candidate Pitch to Roblox

I am applying for the Applied Scientist - AI & World Simulation Intern (2027) role with a commitment to mathematical rigor and software craftsmanship. Having developed autonomous agent runtimes, sub-millisecond search indexes, and verified testing pipelines, I specialize in eliminating systemic operational latency and engineering reliable systems under extreme concurrency. My background in international business operations allows me to evaluate technical trade-offs through the lens of capital efficiency. I look forward to discussing how I can deliver immediate value to Roblox.